

Deep Learning Lab 3

Suraj Karki
Department of Engineering Science
University West
Trollhättan, Sweden
suraj.karki@student.hv.se

Nils Karlsson
Department of Engineering Science
University West
Trollhättan, Sweden
nils.Karlsson@student.hv.se

Nils Wikström
Department of Engineering Science
University West
Trollhättan, Sweden
nils.wikstrom@student.hv.se

Index Terms

Deep Learning, Laboration 3,

Abstract

Playing around with learning rate and seeing how it effects performance of NN

I. TASK 1

Figure X shows the training and validation accuracy and loss of the CNN over 10 epochs. Both training and validation accuracy increase steadily, reaching 98.4% and 99.0%. The training and validation loss decreases without divergence. The validation accuracy remains higher than the training accuracy indicating effective regularization through data augmentation and early stopping, preventing overfitting. The model demonstrates strong generalization performance on the MNIST dataset.

Model Training Performance

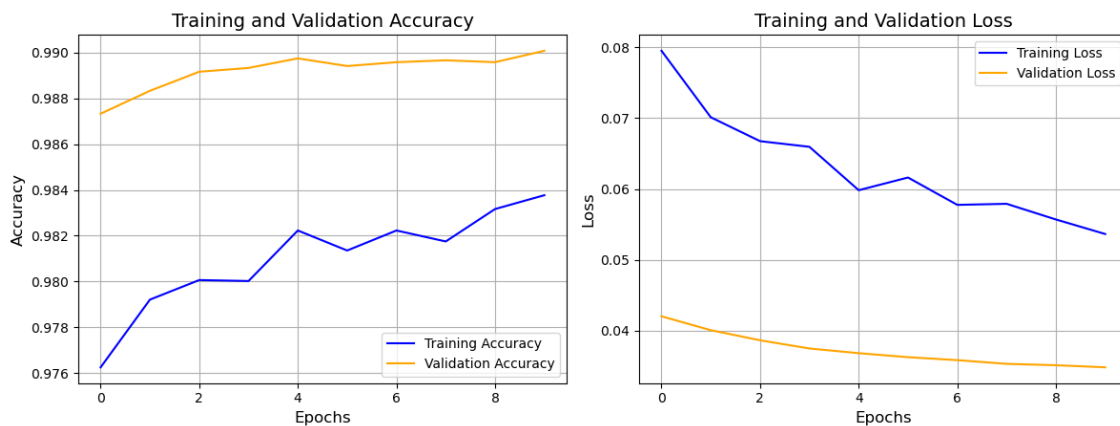


Fig. 1: Model Performance on MNIST dataset

By using Tensorflows Imagegenerator and turning images around or zooming in on them creating more variance of data dropping unnecessary data and implementing early stops made it so a high precision of 98.98% got achieved.

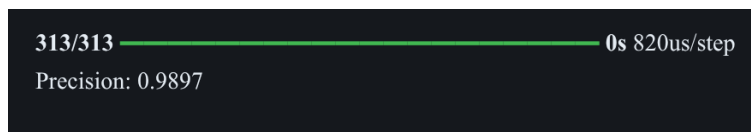


Fig. 2: Precision score

II. TASK 2

test 2

III. TASK 3

test 3